

👉 XAI Lecture 13

Explainable Medical Coding & Attention is not Explanation

FAMAF - UNC

First Semester 2026



eXplainable
Artificial Intelligence

Disclaimer ---

This course is based on

Explainable Artificial Intelligence

From Simple Predictors to Complex Generative Models

Spring 2023, Harvard University

<https://interpretable-ml-class.github.io/>



Explainable Prediction of Medical Codes from Clinical Text

James Mullenbach, Sarah Wiegreffe, Jon Duke, Jimeng Sun, Jacob Eisenstein

Georgia Institute of Technology

{jmullenbach3, swiegreffe6, jon.duke}@gatech.edu

jsun@cc.gatech.edu, jacob@gatech.edu

Abstract

Clinical notes are text documents that are created by clinicians for each patient encounter. They are typically accompanied by medical codes, which describe the diagnosis and treatment. Annotating these codes is labor intensive and error prone; furthermore, the connection between the codes and the text is not annotated, obscuring the reasons and details behind specific diagnoses and treatments. We present

and over 140,000 codes combined in the newer ICD-10-CM and ICD-10-PCS taxonomies ([World Health Organization, 2016](#)). Second, clinical text includes irrelevant information, misspellings and non-standard abbreviations, and a large medical vocabulary. These features combine to make the prediction of ICD codes from clinical notes an especially difficult task, for computers and human coders alike ([Birman-Deych et al., 2005](#)).

In this application paper, we develop convolu-

Figure 1: alt text



Introduction ---

- **Clinical notes:** free text narratives generated by clinicians during patient encounters
- **ICD codes:** metadata codes accompanying clinical notes
 - ▶ Standardized way of indicating diagnoses and procedures
 - ▶ Multiple use cases (billing, predictive modelling)

In Plain English

Given a doctor's note, can a model automatically assign the right diagnostic codes — and point to the exact sentence in the text that justifies each one?

HISTORY

47 year old male with mid-abdominal epigastric pain¹, associated with severe nausea & vomiting; unable to keep down any food or liquid. Pain has become "severe" and constant. Has had an estimated 13 pound weight loss over the past month. Patient reports eating 12 sausages at the Sunday church breakfast five days ago which he believes initiated his symptoms. Patient admits to a history of alcohol dependence². Consuming 5 – 6 beers per day now, down from 10 – 12 per day 6 months ago. States that he has nausea and sweating with "the shakes" when he does not drink.

EXAM

VS: T 99.8°F, otherwise normal.

Mild jaundice noted.

Abdomen distended and tender across upper abdomen³. Guarding is present. Bowel sounds diminished in all four quadrants.

Oral mucosa dry, chapped lips, decreased skin turgor

ASSESSMENT AND PLAN

Dehydration and suspected acute pancreatitis.

Admit to the hospital. Orders written and sent to on-call hospitalist.

1L IV NS started in office. Blood drawn for labs.

Recommend behavioral health counseling for substance abuse assessment and possible treatment.

Patient's wife notified of plan; she will transport to hospital by private vehicle.

ICD-9-CM DIAGNOSIS CODES

789.06 Abdominal pain, epigastric

789.60 Abdominal tenderness, unspecified site

782.4 Jaundice NOS

276.51 Dehydration

303.90 Other and unspecified alcohol dependence, unspecified

ICD-10-CM DIAGNOSIS CODES

R10.13 Epigastric pain

R10.819 Abdominal tenderness, unspecified site

R17 Unspecified jaundice

E86.0 Dehydration

F10.20 Alcohol dependence, uncomplicated

Scenario reference: <https://www.practicefusion.com/icd-10/clinical-concepts-for-family-practice/icd-10-clinical-scenarios/>



Motivation ---

- **Limitation of ICD code annotation**

- ▶ Labor intensive
- ▶ Coders read 720–1,440 pages of clinical documentation per day
- ▶ Prone to error
- ▶ Lack of connection bridging text and code

- **Challenges of automatic coding**

- ▶ Features of clinical text: irrelevant information, misspellings, non-standard abbreviations, large medical vocabulary
- ▶ Label space

Summary of Contributions ---

- **CAML**: Convolutional Attention for Multi-Label classification
 - ▶ Attentional convolutional network: CNN + per-label attention mechanism
 - ▶ Important information may be in short snippets anywhere in the document
 - ▶ Decision support in clinical domain
- **Accuracy Evaluation**
 - ▶ Used open-access MIMIC datasets
 - ▶ Outperformed previously proposed approaches
- **Interpretability Evaluation**
 - ▶ Physician rated the informativeness of automatically generated explanations

🍷 Related Work: Automatic ICD Coding ---

- Structured text v. unstructured text
- **Datasets**
 - ▶ Subset of full ICD label space (Wang et al., 2016)
 - ▶ Subset of medical scenarios (Zhang et al., 2017)
 - ▶ Death certificates in English + French (Névél et al., 2017)
 - ▶ Private datasets (Subotin and Davis, 2016)
- **Approaches**
 - ▶ Recurrent networks with HA-GRU (Baumel et al., 2018)
 - ▶ Character-aware LSTMs (Shi et al., 2017)
 - ▶ Memory networks for top-50/100 codes (Prakash et al., 2017)
 - ▶ Grounded RNN for individual labels (Vani et al., 2017)

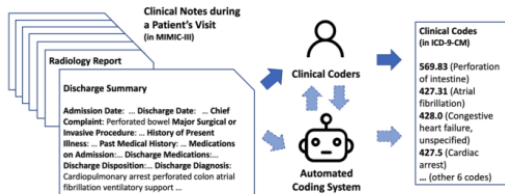


Image reference: <https://www.nature.com/articles/s41746-022-00705-7.pdf>

CAML instead operates on the full ICD-9 label space using MIMIC-III, an open-access dataset.

Related Work: Explainable Text Classification ---

- **Latent rationales** (Lei et al., 2016): learn a latent variable that tags each word as relevant or irrelevant to the predicted label
- **Gradient-based salience** (Li et al., 2016): measure word importance by how much each word affects the output
- **Attention as explanation** (Rush et al., 2015; Rocktäschel et al., 2016): highlight salient text features using attention weights

Key Gap

Prior work identifies *which* words the model focuses on, but never verifies whether **attending to completely different words** would have changed the prediction — leaving the causal link between attention and output unexamined.

🍲 Related Work: Attentional Convolution for NLP ---

- CNNs applied to sentiment classification (Kim, 2014) and language modeling
- Convolution and attention: Allamanis et al., 2016; Yin et al., 2016; dos Santos et al., 2016; Yin and Schütze, 2017
- Most relevant work (Yang et al., 2016 and Allamanis et al., 2016):
 - ▶ Use context vectors (one for document) to compute attention over specific locations in the text

CAML Differentiation

- Goal: select locations most important for predicting specific labels
- Compute separate attention weights for each label in the label space

🍲 ICD Code Prediction Problem ---

Clinical Text

A 12-year old girl with known hyperagglutinability, presented to the emergency department with a 2-week history of headaches and facial weakness. Neurologic examination indicated sensorineural hearing loss on the right side with Weber's test lateralizing to the left, and the Rinne's test demonstrating bone conduction greater than air conduction on the right. Magnetic resonance imaging of the head revealed severe structural defects of the right petrous temporal bone. No indication of cerebral infarction.

ICD Codes

Diagnosis	ICD-9	ICD-10
Cervical Sprain, initial encounter	847.0	S13.4xxA
Thoracic Sprain, initial encounter	847.1	S23.3xxA
Lumbar Sprain, initial encounter	847.2	S33.5xxA
Cervical Degenerative Disc Disease	722.4	M50
Thoracic Degenerative Disc Disease	722.51	M51
Lumbar Degenerative Disc Disease	722.52	M51.2

Multi-Label Classification Problem

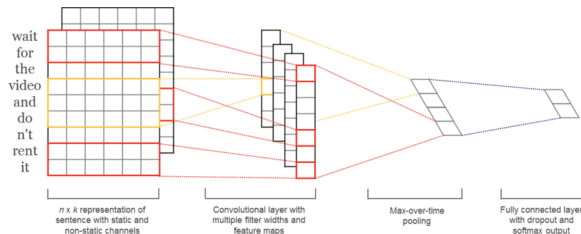
- “Match” snippets of clinical text to a label — each code is triggered by a **different part** of the document, not the document as a whole
- For rare codes with few training examples, use the **code's own text description** as an extra learning signal (e.g., “Cervical Degenerative Disc Disease” tells the model what kind of text should activate it)



“IT’S
NEVER
LUPUS.”

peacock  **H O U S E** M.D.

🍷 Typical Convolutional N-Gram Model ---

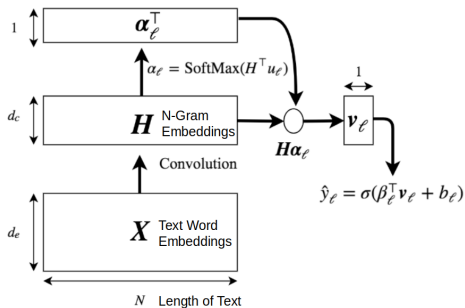


- Each word is represented as a dense vector (**word embedding**)
- A convolutional filter slides over **N consecutive words** at a time, detecting local patterns (e.g., “acute renal failure”)
- Each filter learns to recognize a specific “**word snippet**” motif
- **Max pooling** asks: *was this pattern detected anywhere in the document?* — collapsing position information in the process

Key Limitation

Max pooling discards *where* the pattern appeared — you know a pattern was detected, but you can't point to the sentence that triggered it, making the prediction unexplainable to a physician.

Convolutional Attention Model ---



Key Insight

Because $\mathbf{u}_\ell$ is different for every label, each code gets its own attention distribution — and therefore its own extractable text snippet as explanation.

1 Convolution: $X \rightarrow H$

Each column of H is an n -gram embedding — a single vector capturing the meaning of N consecutive words in context.

2 Per-label attention: $\alpha_\ell = \text{SoftMax}(H^\top \mathbf{u}_\ell)$

For each label ℓ , compare every n -gram against the label vector $\mathbf{u}_\ell$ via dot product, then normalize. Result: a distribution over document positions — *which fragment is most relevant for this code?*

3 Weighted aggregation: $\mathbf{v}_\ell = \sum_{n=1}^N \alpha_{\ell,n} \mathbf{h}_n$

Instead of max pooling, sum all n -grams weighted by attention — preserving *where* the evidence came from.

4 Classification: $\hat{y}_\ell = \sigma(\beta_\ell^\top \mathbf{v}_\ell + b_\ell)$

Linear layer + sigmoid gives the probability for each code independently.

🍷 Training Loss - Binary Cross-Entropy (CAML) ---

$$L_{\text{BCE}}(\mathbf{X}, y) = - \sum_{\ell=1}^{\mathcal{L}} [y_{\ell} \log(\hat{y}_{\ell}) + (1 - y_{\ell}) \log(1 - \hat{y}_{\ell})]$$

In Plain English

Each code is treated as an **independent binary classification** — present or not — and errors are accumulated over all ~9,000 possible codes.

🍷 Training Loss Function - With Description Regularization (DR-CAML) 2/2 ---

$$L(\mathbf{X}, y) = L_{\text{BCE}} + \lambda \frac{1}{n_y} \sum_{\ell: y_\ell=1} \|\mathbf{z}_\ell - \beta_\ell\|_2$$

- n_y : number of true labels for data example
- $\mathbf{z}_\ell$: representation of label based on code description
- β_ℓ : representation of label learned by model

In Plain English

For common codes, learn from data. For rare codes, stay close to what the medical dictionary says the code means.

Experimental Setup ---

Dataset: MIMIC-III

- **Input:** discharge summary (max 2,500 tokens)
- **Output:** ICD-9 codes — 7,000 diagnosis + 2,000 procedure (~9,000 total)
- **Embeddings:** 100-dim word2vec pretrained on corpus

	MIMIC-III full
# training documents	47,724
Vocabulary size	51,917
Mean tokens / document	1,485
Mean labels / document	15.9
Total labels	8,922

Baselines

- Single-layer CNN (Kim, 2014)
- Logistic Regression (bag-of-words)
- Bidirectional GRU (BiGRU)

Evaluation Metrics

- Macro / Micro AUC and F1
- Precision@ n — fraction of top- n predicted codes that are correct

Experimental Setup - Resume

Each document is a long medical narrative; the model must simultaneously predict ~16 codes on average, chosen from a space of nearly 9,000 — making this an extremely sparse multi-label problem.

🍲 Results 1/2 ---

Model	AUC		F1				P@n	
	Macro	Micro	Macro	Micro	Diag	Proc	8	15
Scheurwegs et. al (2017)	–	–	–	–	0.428	0.555	–	–
Logistic Regression	0.561	0.937	0.011	0.272	0.242	0.398	0.542	0.411
CNN	0.806	0.969	0.042	0.419	0.402	0.491	0.581	0.443
Bi-GRU	0.822	0.971	0.038	0.417	0.393	0.514	0.585	0.445
CAML	0.895	0.986*	0.088	0.539*	0.524*	0.609*	0.709*	0.561*
DR-CAML	0.897	0.985	0.086	0.529	0.515	0.595	0.690	0.548

- CAML gives strongest results on all metrics; DR-CAML is the regularized version ($\lambda = 0.01$)
- Bi-GRU comparable to vanilla CNN; Logistic Regression worse than all neural architectures

🍷 Results - Micro vs. Macro Metrics ---

Model	AUC		F1				P@n	
	Macro	Micro	Macro	Micro	Diag	Proc	8	15
Scheurwegs et. al (2017)	—	—	—	—	0.428	0.555	—	—
Logistic Regression	0.561	0.937	0.011	0.272	0.242	0.398	0.542	0.411
CNN	0.806	0.969	0.042	0.419	0.402	0.491	0.581	0.443
Bi-GRU	0.822	0.971	0.038	0.417	0.393	0.514	0.585	0.445
CAML	0.895	0.986*	0.088	0.539*	0.524*	0.609*	0.709*	0.561*
DR-CAML	0.897	0.985	0.086	0.529	0.515	0.595	0.690	0.548

- **Macro:** computes the metric separately for each code then averages — every code counts equally, regardless of frequency. A very rare code weighs the same as a very common one.
- **Micro:** pools all true positives, false positives, and false negatives across all codes before computing — frequent codes dominate the result.

In this problem the difference is critical

A model that ignores rare codes can still achieve high Micro-F1, but will be heavily penalized by Macro-F1.

🍷 Interpretability Evaluation ---

- Sample of 100 predicted codes
- Most important k -gram ($k = 4$) + five words on each side for context
- Asked the Physician how informative it is

🍷 Another Baseline: Cosine Similarity Explanation ---

One of the baselines for extracting the text snippet that explains each code.

- Given an ICD code, it computes the cosine similarity between each 4-gram in the document and the code's text description — and returns the most similar one.

It is a purely lexical method: no model involved, just word comparison.

Key Limitation

Works well when the document **literally repeats** words from the code description — **fails** when the relationship is implicit or uses different terminology.



Interpretability: Qualitative Examples ---

934.1: "Foreign body in main bronchus"

CAML (HI)	<i>...line placed bronchoscopy performed showing large mucus plug on the left on transfer to...</i>
Cosine Sim	<i>...also needed medication to help your body maintain your blood pressure after receiving iv...</i>
CNN	<i>...found to have a large III lingular pneumonia on chest x ray he was...</i>
Logistic Regression	<i>...impression confluent consolidation involving nearly the entire left lung with either broncho-centric or vascular...</i>

442.84: "Aneurysm of other visceral artery"

CAML (I)	<i>...and gelfoam embolization of right hepatic artery branch pseudoaneurysm coil embolization of the gastroduodenal...</i>
Cosine Sim	<i>...coil embolization of the gastroduodenal artery history of present illness the pt is a...</i>
CNN	<i>...foley for hemodynamic monitoring and serial hematocrits angio was performed and his gda was...</i>
Logistic Regression (I)	<i>...and gelfoam embolization of right hepatic artery branch pseudoaneurysm coil embolization of the gastroduodenal...</i>

428.20: "Systolic heart failure, unspecified"

CAML	<i>...no mitral valve prolapse moderate to severe mitral regurgitation is seen the tricuspid valve...</i>
Cosine Sim	<i>...is seen the estimated pulmonary artery systolic pressure is normal there is no pericardial...</i>
CNN	<i>...and suggested starting hydralazine imdur continue aspirin arg admitted at baseline cr appears patient...</i>
Logistic Regression (HI)	<i>...anticoagulation monitored on tele pump systolic dysfunction with ef of seen on recent echo...</i>

🍷 How to Pick the Most Influential 4-gram? ---

- **CAML**: looks at SoftMax output α_ℓ to get the best 4-gram for code ℓ
- **Logistic Regression**: sum of weight matrix coefficients for ℓ over words in k -gram; top-scoring k -gram returned
- **Code descriptions**: cosine similarity between each stemmed k -gram and stemmed ICD-9 code description
- **Max-Pooling CNN**:

$$a_i = \arg \max_{j \in \{1, \dots, m-k+1\}} (H_{ij}), \quad \alpha_{i\ell} = \sum_{j: a_j=i} \beta_{\ell,j}$$

Select the most important k -gram as $\arg \max_i \alpha_{i\ell}$

Qualitative Evaluation Results ---

Method	Informative	Highly informative
CAML	46	22
Code Descriptions	48	20
Logistic Regression	41	18
CNN	36	13

- CAML selects the greatest number of “highly informative” explanations
- CAML selects more “informative” explanations than both CNN and logistic regression
- Cosine Similarity also performs well; examples in Table 1 demonstrate strengths of CAML

Conclusion ---

Concluding Thoughts

- Yields strong improvements over previous metrics across several formulations of ICD-9 code prediction
- Provides satisfactory explanations for its predictions per stakeholder evaluation

Future Work

- **Extensible** to other multi-label document tagging tasks
- **Linguistic perspective**: integrate document structure of discharge summaries; better handle out-of-vocabulary tokens
- **Application perspective**: leverage hierarchy of ICD codes; predict codes for future visits

Discussion (Paper 1) ---

- ① What are potential limitations of CAML?
- ② The explanations from the attention mechanism are extracted text snippets. Does this provide enough information? What else could be provided?
- ③ Strengths and weaknesses of their evaluation and results (accuracy and interpretability)
- ④ What additional applications could CAML be useful for?

Attention is not Explanation

Sarthak Jain
Northeastern University
jain.sar@husky.neu.edu

Byron C. Wallace
Northeastern University
b.wallace@northeastern.edu

Abstract

Attention mechanisms have seen wide adoption in neural NLP models. In addition to improving predictive performance, these are often touted as affording transparency: models equipped with attention provide a distribution over attended-to input units, and this is often presented (at least implicitly) as communicating the relative importance of inputs. However, it is unclear what relationship ex-

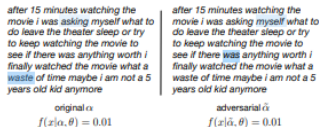


Figure 1: Heatmap of attention weights induced over a negative movie review. We show observed model attention (left) and an adversarially constructed set of at-

(Jain and Wallace 2019)



Attention Mechanism: a "Recent" Breakthrough in NLP ---

- **2015**: birth of attention mechanism in NLP
 - ▶ “Neural machine translation by jointly learning to align and translate” (Bahdanau et al., 2015)
- **2017**: birth of transformers based on self-attention mechanism
 - ▶ “Attention is all you need” (Vaswani et al., 2017)

Task: Hotel service

you get what you pay for . not the cleanest rooms but bed was **clean** and so was bathroom . bring your own **towels** though as very **thin** . **service was excellent** ! let us book in at 8:30am ! for location and price , this ca n't be beaten , but it is **cheap** for a reason . if you come expecting the hilton , then book the hilton ! for uk travellers , think of a blackpool b&b.

Fig. 1. Example of attention visualization for an aspect-based sentiment analysis task, from [1, Fig. 6]. Words are highlighted according to attention scores. Phrases in bold are the words considered relevant for the task or human rationales.



Many Use Attention as an Explanation Mechanism ---

Show, Attend and Tell: Neural Image Caption Generation with Visual Attention

Kelvin Xu
Jimmy Lei Ba
Ryan Kiros
Kyunghyun Cho
Aaron Courville
Ruslan Salakhutdinov
Richard S. Zemel
Yoshua Bengio

KELVIN.
JIMM*
RKIRC
KYUNGHYUN.C
AARON.COURVILLE
RSALAKH

Chen et al. *BMC Medical Informatics and Decision Making* 2020, **20**(Suppl 3):131
<https://doi.org/10.1186/s12911-020-1110-7>

RETAIN: An Interpretable Predictive Model for Healthcare using Reverse Time Attention Mechanism

Edward Choi*, Mohammad Taha Bahadori*, Joshua A. Kulas*,
Andy Schuetz†, Walter F. Stewart†, Jimeng Sun*
* Georgia Institute of Technology † Sutter Health
imn2892.bahadori@kulas21@gatech.edu,
j.kulas@gatech.edu

BMC Medical Informatics and
Decision Making

gatech.edu

RESEARCH

Open Access

Interpretable clinical prediction via attention-based neural network



Peipei Chen^{1,2}, Wei Dong³, Jinliang Wang⁴, Xudong Lu^{1,2}, Uzay Kaymak^{2,1} and Zhengxing Huang^{1*}

From 5th China Health Information Processing Conference
Guangzhou, China. 22-24 November 2019

Attention-based explanations are widespread across NLP and high-stakes domains like clinical prediction — making it critical to ask whether they are actually faithful.



But Does Attention Provide Faithful Explanations? ---

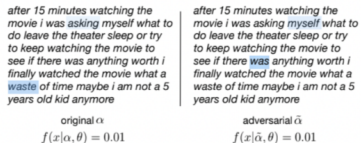


Figure 1: Heatmap of attention weights induced over a negative movie review. We show observed model attention (left) and an adversarially constructed set of attention weights (right). Despite being quite dissimilar, these both yield effectively the same prediction (0.01).

The two heatmaps are **very different** — yet both yield the same prediction.

Original α	Adversarial $\tilde{\alpha}$
Highlights “waste” $f(x \alpha, \theta) = 0.01$	Highlights “was” $f(x \tilde{\alpha}, \theta) = 0.01$

In Plain English

If attending to completely different words produces the exact same prediction, then attention cannot be explaining *why* the model made that decision — it is just one of many equivalent weight configurations.

Research Questions ---

To what degree do attention weights provide meaningful explanations for predictions?

- ① **Consistency:** Do attention weights **correlate with other feature importance measures**, specifically gradients and leave-one-out (LOO)?
 - ▶ If attention is a faithful explanation, it should agree with methods that have clearer causal semantics.
- ② **Counterfactual attention:** Would **alternative attention distributions** necessarily yield different predictions?
 - ▶ If the model predicts the same thing regardless of which words are attended to, attention cannot be the cause of the prediction.

🍷 Contributions and Main Findings ---

- ① Standard attention modules **do not provide meaningful explanations** and should not be treated as though they do.
- ② Learned attention weights are frequently **uncorrelated with gradient-based** measures of feature importance.
- ③ One can identify very **different attention distributions** that nonetheless yield equivalent predictions.

🍷 Experimental Setup: Datasets ---

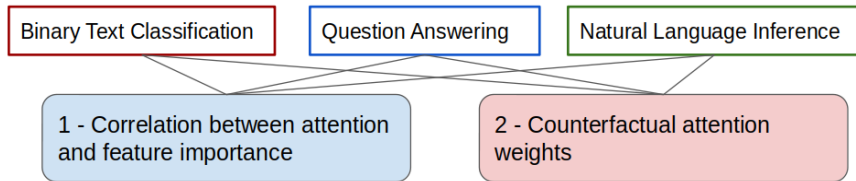
<i>Dataset</i>	<i> V </i>	<i>Avg. length</i>	<i>Train size</i>	<i>Test size</i>	<i>Test performance (LSTM)</i>
SST	16175	19	3034 / 3321	863 / 862	0.81
IMDB	13916	179	12500 / 12500	2184 / 2172	0.88
ADR Tweets	8686	20	14446 / 1939	3636 / 487	0.61
20 Newsgroups	8853	115	716 / 710	151 / 183	0.94
AG News	14752	36	30000 / 30000	1900 / 1900	0.96
Diabetes (MIMIC)	22316	1858	6381 / 1353	1295 / 319	0.79
Anemia (MIMIC)	19743	2188	1847 / 3251	460 / 802	0.92
CNN	74790	761	380298	3198	0.64
bAbI (Task 1 / 2 / 3)	40	8 / 67 / 421	10000	1000	1.0 / 0.48 / 0.62
SNLI	20982	14	182764 / 183187 / 183416	3219 / 3237 / 3368	0.78

The two experiments run across **three NLP task types** and **11 datasets**, ensuring conclusions are not specific to any single domain.

- **Binary Text Classification** (red): sentiment, medical records, news — varying document lengths from 19 to 2,188 tokens
- **Question Answering** (blue): CNN news articles and bAbI tasks
- **Natural Language Inference** (green): SNLI — 570k sentence pairs

All models use a **BiLSTM encoder** with standard additive attention.

🍷 Experimental Setup: Tasks ---



Each dataset is used to run both experiments:

- ① **Correlation:** do attention weights agree with gradient and LOO importance scores across all task types?
- ② **Counterfactual:** can we find alternative attention distributions that preserve the prediction — across classification, QA, and NLI?

Key Insight

Testing across 3 task types and 11 datasets rules out task-specific artifacts — if attention fails as explanation everywhere, the finding is general.

🍷 Correlation Between Attention and Feature Importance ---

To what extent do attention weights agree with other measures of feature importance?

Two reference methods are used as ground truth for comparison:

- ① **Feature gradients:** how much does the output change if we perturb each input word?
Measures sensitivity of the prediction to each feature.
- ② **Leave-One-Out (LOO):** remove one word at a time and measure the drop in prediction.
Directly tests the causal impact of each word.

Key Insight

If attention is a faithful explanation, it should **rank words similarly** to these methods, both of which have clearer causal semantics than attention.

Distribution Change Measure- Total Variation Distance (TVD) ---

TVD measures how different two output distributions are after changing attention.

$$\text{TVD}(\hat{y}_1, \hat{y}_2) = \frac{1}{2} \sum_{i=1}^{|\mathcal{Y}|} |\hat{y}_{1i} - \hat{y}_{2i}|$$

In Plain English

TVD close to 0 means the prediction barely changed despite different attention. If we swap attention weights and TVD stays near 0, attention was not driving the output.

🍷 Distribution Change Measure - Kendall's τ coefficient ---

Kendall's τ measures how well two importance rankings agree.

$$\tau = \frac{(\text{concordant pairs}) - (\text{discordant pairs})}{(\text{number of pairs})} = 1 - \frac{2(\text{discordant pairs})}{\binom{n}{2}}$$

In Plain English

$\tau = 1$ means both methods rank words identically. $\tau = 0$ means no agreement. If attention and gradients disagree on word ranking, they are not measuring the same thing.

Algorithm for Feature Importance Computations ---

- ① $\mathbf{h} \leftarrow \text{Enc}(\mathbf{x}), \quad \hat{\alpha} \leftarrow \text{softmax}(\phi(\mathbf{h}, \mathbf{Q}))$
- ② $\hat{y} \leftarrow \text{Dec}(\mathbf{h}, \alpha)$
- ③ $g_t \leftarrow \left| \sum_{w=1}^{|V|} \mathbf{1}[\mathbf{x}_{tw} = 1] \frac{\partial y}{\partial \mathbf{x}_{tw}} \right|, \quad \forall t \in [1, T]$
- ④ $\tau_g \leftarrow \text{Kendall-}\tau(\alpha, g)$
- ⑤ $\Delta \hat{y}_t \leftarrow \text{TVD}(\hat{y}(\mathbf{x}_{-t}), \hat{y}(\mathbf{x})), \quad \forall t \in [1, T]$
- ⑥ $\tau_{\text{loo}} \leftarrow \text{Kendall-}\tau(\alpha, \Delta \hat{y})$

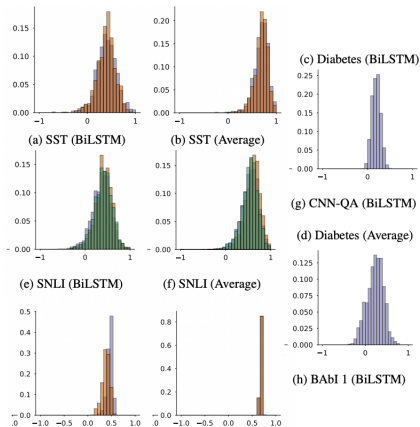
- ① **Encode and attend**
- ② **Decode prediction**
- ③ **Gradient importance:** How much does the output change if we perturb word t ?
- ④ **Correlation with attention**
- ⑤ **LOO importance** for each token t : How much does the output change if we remove word t ?
- ⑥ ****Correlation with attention**

In Plain English

Compute two independent word rankings (gradient, LOO) and measure how well attention agrees with each. Low τ means attention is ranking words differently from methods with clearer causal meaning.

🍷 Feature Gradients Results ---

Histograms of τ_g (Kendall correlation between attention and gradients) across all test examples.



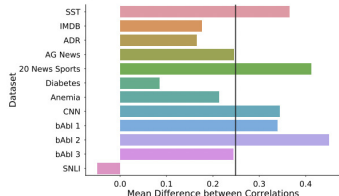
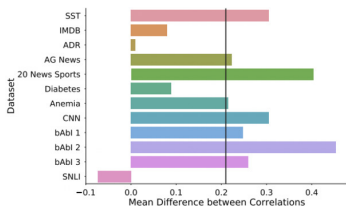
- **BiLSTM:** centered around 0.2–0.4 with high variance — weak and inconsistent agreement.
- **Average encoder:** shifts right and concentrates — attention does reflect importance.

Key Insight

The problem is not attention itself — it is the combination with BiLSTM. Hidden states mix information from all words, so attention weights no longer correspond directly to the importance of individual input tokens.

🍷 Correlation Comparison: Attention, Gradients, L00 ---

Both plots show the **mean difference in correlation** across datasets. Positive values mean gradients/LOO agree more with each other than with attention.



- **Top:** L00 vs. Gradients minus Attention vs. L00 — gradients and L00 agree with each other $\sim 0.2 \tau$ more than attention agrees with L00.
- **Bottom:** L00 vs. Gradients minus Attention vs. Gradients — same pattern, $\sim 0.25 \tau$ gap on average.

Key Insight

Gradients and L00 consistently agree with each other more than either agrees with attention. **Attention is measuring something different from feature importance.**

🍲 Possible Limitations in Experiment Results ---

- ➊ Potential issues with Kendall τ correlation — many irrelevant features may add noise to the correlation measures
- ➋ Alternative measures of feature importance (gradients and LOO) may not be considered ideal or the 'ground truth' for comparison purposes

🍷 Counterfactual Attention Weights ---

If attention explains predictions, then changing attention should change the prediction.

- ① **Consistency**: correlation with other measures of feature importance
- ② **Counterfactual attention distributions** should yield corresponding changes in predictions

brilliant	and	moving	performances	by	tom	and	peter	finch
brilliant	and	moving	performances	by	tom	and	peter	finch
brilliant	and	moving	performances	by	tom	and	peter	finch

🍷 Counterfactual Attention: Two Experiments ---

- ① **Attention Permutation:** scramble the original attention weights
- ② **Adversarial Attention:** generate an adversarial distribution that is maximally distinct from the observed attention weights, whilst simultaneously close to the observed prediction

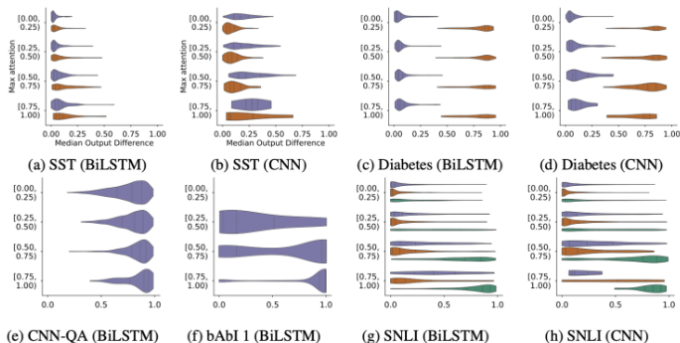
Attention Permutation: Algorithm ---

```
h  $\leftarrow$  Enc(x),  $\hat{\alpha} \leftarrow \text{softmax}(\phi(\mathbf{h}, \mathbf{Q}))$   
 $\hat{y} \leftarrow \text{Dec}(\mathbf{h}, \hat{\alpha})$   
for  $p \leftarrow 1$  to 100 do  
   $\alpha^p \leftarrow \text{Permute}(\hat{\alpha})$   
   $\hat{y}^p \leftarrow \text{Dec}(\mathbf{h}, \alpha^p)$   $\triangleright$  h not changed  
   $\Delta \hat{y}^p \leftarrow \text{TVD}[\hat{y}^p, \hat{y}]$   
end for  
 $\Delta \hat{y}^{med} \leftarrow \text{Median}_p(\Delta \hat{y}^p)$ 
```

- **h** (the hidden state) is kept fixed
- Only the attention weights are permuted
- Measures how much the output changes when attention is scrambled

🍷 Attention Permutation: Results ---

Each violin shows the distribution of $\Delta \hat{y}^{med}$ (change in output) after **randomly permuting** attention weights, grouped by max attention value.



Key Insight

Permuting attention barely changes predictions in most tasks, but does matter when the task requires identifying a specific token (Diabetes, CNN-QA, bAbI).

Adversarial Attention ---

Explicitly seek out attention weights that differ as much as possible from observed attention weights, yet leaving prediction almost unchanged.

Why?

- Alternative attention distributions identified for the same output may be viewed as equally plausible explanations for the same output. **Is this undesirable ?**
 - ▶ Cannot conclude that model made a specific prediction because it attended over inputs in a certain way.
 - ▶ Lack of specificity makes constructing counterfactual explanations hard?
 - ▶ Makes model less trustworthy?



Adversarial Attention: Optimisation ---

Seek $\alpha^{(1)}, \alpha^{(2)}, \alpha^{(3)} \dots \alpha^{(k)}$.

ϵ - Quantifies small difference in model output

Measure of distance between attention distributions ?

$$JSD(\alpha_1, \alpha_2) = \frac{1}{2} (KL[\alpha_1 | \frac{\alpha_1 + \alpha_2}{2}] + KL[\alpha_2 | \frac{\alpha_1 + \alpha_2}{2}])$$

$$\underset{\alpha^{(1)}, \dots, \alpha^{(k)}}{\text{maximize}} \quad f(\{\alpha^{(i)}\}_{i=1}^k)$$

$$\text{subject to} \quad \forall i \text{ TVD}[\hat{y}(\mathbf{x}, \alpha^{(i)}), \hat{y}(\mathbf{x}, \hat{\alpha})] \leq \epsilon \quad (1)$$

Where $f(\{\alpha^{(i)}\}_{i=1}^k)$ is:

$$\sum_{i=1}^k JSD[\alpha^{(i)}, \hat{\alpha}] + \frac{1}{k(k-1)} \sum_{i < j} JSD[\alpha^{(i)}, \alpha^{(j)}] \quad (2)$$

In Plain English

Find attention maps that are as different as possible from the original and from each other — while the model still predicts the same thing.

Adversarial Attention: Algorithm ---

$\mathbf{h} \leftarrow \text{Enc}(\mathbf{x}), \hat{\alpha} \leftarrow \text{softmax}(\phi(\mathbf{h}, \mathbf{Q}))$

$\hat{y} \leftarrow \text{Dec}(\mathbf{h}, \hat{\alpha})$

$\alpha^{(1)}, \dots, \alpha^{(k)} \leftarrow \text{Optimize Eq 1}$

for $i \leftarrow 1$ to k **do**

$\hat{y}^{(i)} \leftarrow \text{Dec}(\mathbf{h}, \alpha^{(i)})$ $\triangleright \mathbf{h}$ is not
 changed

$\Delta \hat{y}^{(i)} \leftarrow \text{TVD}[\hat{y}, \hat{y}^{(i)}]$

$\Delta \alpha^{(i)} \leftarrow \text{JSD}[\hat{\alpha}, \alpha^{(i)}]$

end for

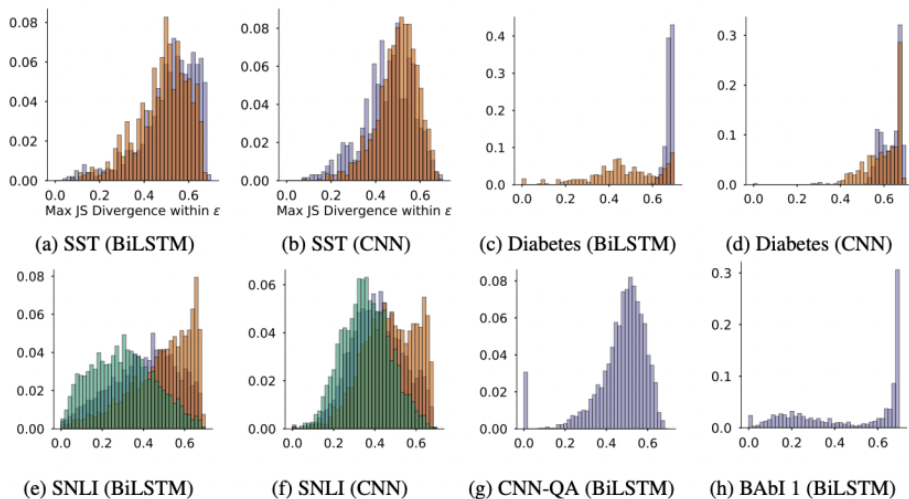
$\epsilon\text{-max JSD} \leftarrow \max_i \mathbf{1}[\Delta \hat{y}^{(i)} \leq \epsilon] \Delta \alpha^{(i)}$

- 1 Encode input and compute original $\hat{\alpha}$ and $\hat{y}$.
- 2 Find k adversarial distributions by solving the optimisation problem.
- 3 Decode each using the **same hidden states $\mathbf{h}$** — only attention changes.
- 4 Report ϵ -max JSD: the largest attention divergence among distributions that kept the prediction within ϵ of the original.

In Plain English

ϵ -max JSD tells you how different attention can be while the model still predicts the same thing — the higher it is, the less faithful attention is.

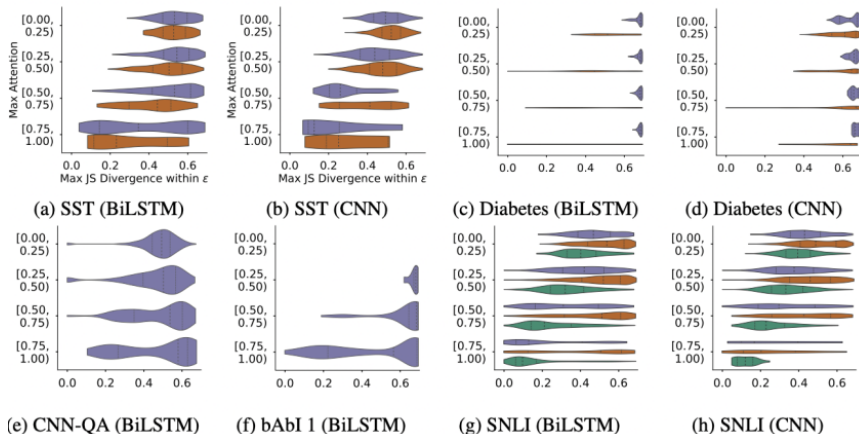
🍷 Adversarial Attention: Results ---



All histograms are skewed to the right.



Adversarial Attention: Results ---



For most tasks, you can construct attention distributions that are maximally different from the original and still get the same prediction — attention is not uniquely determining the output.

🍲 Counterfactual Attention: Limitations ---

- Only standard additive and dot-product attention are tested — alternate mechanisms may behave differently.
- Evaluations are limited to classification, QA, and NLI tasks — conclusions may not extend to seq2seq tasks like machine translation.

Overall Limitations ---

- Alternate attention mechanisms are not explored
- Limited evaluations to classification, QA, and NPI tasks; seq2seq tasks are not considered

🍷 Critique and Discussion 1/2 ---

Wiegreffe & Pinter (2019) — “**Attention is not not Explanation**” directly respond to Jain & Wallace, raising three key objections:

- Attention distribution is not a primitive — the model can redistribute information through hidden states before attention is applied.
- Is attention *necessary* for prediction, or just one of many compatible configurations?
- How adversarial are the adversaries? — the constructed distributions may not be realistic alternatives.

🍷 Critique and Discussion 2/2 ---

Other relevant follow-ups:

- Serrano & Smith (2019): removing high-attention weights does not always change predictions.
- Bastings & Filippova (2020): the core problem is the lack of consensus on what “faithful explanation” even means.

Open questions for discussion:

- Do you agree with the authors' conditions for a faithful explanation?
- Is attention a *sufficient* explanation, even if not a complete one?
- What other conditions should a good explanation satisfy?
- How can domain experts be brought in to evaluate explanation quality?

Thank You!

References --- |

- Jain, Sarthak, and Byron C. Wallace. 2019. “Attention Is Not Explanation.” In *Proceedings of NaacI-Hlt*, 3543–56.
- Mullenbach, James, Sarah Wiegrefte, Jon Duke, Jimeng Sun, and Jacob Eisenstein. 2018. “Explainable Prediction of Medical Codes from Clinical Text.” *arXiv Preprint arXiv:1802.05695*.